Forensic Science International: Genetics 85 (2026) 103527

Contents lists available at ScienceDirect

Forensic Science International: Genetics

journal homepage: www.elsevier.com/locate/fsigen

DNA tests for ancestry and phenotype inference applied to the UK Metropolitan Police Operation Minstead: The investigation of serial sex offender Delroy Grant

C. Phillips [a,b], A. Rodriguez [a], A. Gómez-Tato [c], J. Álvarez-Dios [c], M. Casares de Cal [c], A. Salas [a,d], Á. Carracedo [a,e,f], M.V. Lareu [a,*]

[a] Forensic Genetics Unit, Institute of Forensic Sciences, University of Santiago de Compostela, Santiago de Compostela, Spain
[b] King's Forensics, Faculty of Life Sciences and Medicine, King's College, London, UK
[c] Faculty of Mathematics, University of Santiago de Compostela, Santiago de Compostela, Spain
[d] Unidade de Xenética, Instituto de Ciencias Forenses, Facultade de Medicina, Universidade de Santiago de Compostela, and Genética de Poblaciones en Biomedicina (GenPoB) Research Group, Instituto de Investigación Sanitaria (IDIS), 15706 Hospital Clínico Universitario de Santiago (SERGAS), Spain
[e] Fundacion Pública Galega de Medicina Xenomica (SERGAS), National Genotyping Center (CEGEN), Health Research Institute of Santiago de Compostela (IDIS), Santiago de Compostela, Spain
[f] Genomic Medicine Group-Center for Research in Molecular Medicine and Chronic Diseases (CIMUS), University of Santiago de Compostela, Spain

ARTICLE INFO

Keywords:
Ancestry analysis
Forensic DNA phenotyping
Operation Minstead
Criminal casework
Population admixture
Co-ancestry estimation

ABSTRACT

SNP analysis in forensic genetics has expanded substantially over the past two decades, particularly for ancestry inference and forensic DNA phenotyping (FDP). However, interpretation of data from such analyses in complex, admixed populations is challenging and, if not carefully contextualized, risks misdirecting criminal investigations. Here we revisit Operation Minstead, a major UK criminal investigation leading to the conviction of serial sex offender Delroy Grant in 2011. In describing our ancestry and FDP analyses of Grant's DNA, we critically examine the role and limitations of early forensic SNP analyses in an operational context. Although DNA was readily available, SNP-based ancestry and phenotyping analyses did not meaningfully advance the investigation. We review the population genetics inferences made by different parties between 2004 and 2008, including reports from a commercial ancestry testing company and independent population genetics expertise, and contrast them with analyses performed by the Forensic Genetics Unit, University of Santiago de Compostela. Our evaluation highlights key methodological issues, including overinterpretation of the inferred co-ancestry proportions of the Minstead suspect, lack of transparency in proprietary SNP panels and reference population data used, and insufficient consideration of within-group variance in admixed populations. We discuss the limitations for inferring common and rare pigmentation patterns in early FDP analyses in individuals with predominantly African genomic backgrounds, and the challenges posed by incomplete marker coverage and limited biological understanding of the expression of pigmentation phenotypes at the time. This case illustrates the risks associated with excessive geographic precision in ancestry inference, especially when likelihoods are modest, and overlap of population variation is substantial. We emphasize the importance of combining information from uniparental markers, X-SNPs, and carefully selected ancestry-informative markers with extreme allele frequency differences, plus the value of likelihood-based frameworks over categorical interpretations. Finally, we discuss how lessons learned from Operation Minstead have informed the development, validation, and interpretation of forensic SNP panels in subsequent years. Overall, this retrospective analysis highlights the need for caution, transparency, and statistical rigor in the forensic application of SNP-based ancestry and phenotyping tests, as well as providing guidance for their careful use in current and future criminal investigations.

* Corresponding author.
E-mail address: mvictoria.lareu@usc.es (M.V. Lareu).

https://doi.org/10.1016/j.fsigen.2026.103527
Received 31 December 2025; Received in revised form 1 May 2026; Accepted 18 May 2026
Available online 27 May 2026
1872-4973/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

1. Introduction

Delroy Grant was amongst the most prolific serial sex offenders in UK history, thought to have started his attacks in London in 1990. Five months before his arrest in 2009, it was confirmed he had committed at least 203 offences, a large proportion of them sexual assaults, often very violent attacks, in the homes of elderly victims, including some men. Supplementary File S1 details the timelines of Grant's 17-year period of criminal activity and subsequent police investigations, together with the geographic distribution of offences clustered around eight south London districts. Routine DNA profiling linked the crimes, and Y-STR and mitochondrial analysis was added, but no database hit or link to numerous putative suspect samples was ever established, so specialist DNA analysis was instigated in 2004 to infer the likely ancestry of Grant. Grant's distinctive *modus operandi* of breaking into houses at night, cutting phone-lines and electricity, wearing gloves and balaclava, and attacking from behind, meant victims, often with limited visual acuity, gave eyewitness descriptions lacking sufficient detail to be informative in defining the suspect's appearance. Due to major oversights by investigators, the opportunity to take a sample from Grant was missed on two occasions, because someone with the same name had been previously eliminated by DNA. A notable feature of the pattern of attacks was three lengthy gaps in offences for five years, three years and finally 18 months - suggesting to investigators that Grant was possibly out of the country during this time.

In 2004, the US DNA testing company *DNAprint* offered to analyse Minstead DNA samples to infer the suspect's ancestry. In 2008, the Forensic Genetics Unit at the University of Santiago de Compostela (herein USC) was asked by the Senior Investigating Officer (SIO) to evaluate the ancestry inferences made by *DNAprint* and an independent population genetics expert; to complement these analyses with USC's own SNP tests; and to conduct forensic DNA phenotyping (FDP) tests to analyse the suspect's possible pigmentation traits. This paper describes the challenges of evaluating ancestry analyses made by other parties when their SNP and population data was a 'black box'; the difficulties of inferring skin colour at a time when this field of FDP was still in the early stages of development; and the limitations of evaluating patterns of red hair colour and pale skin phenotypes without full coverage of *MC1R* SNPs. Our report to investigators concluded that Grant was African Caribbean or African American. However, the report's key message stated that *DNAprint* SNP genotypes had inflated the estimates of Grant's Native American co-ancestry proportions, while analyses of this data had produced an unduly precise, unsubstantiated inference of Grant's parental origins in the Caribbean Windward Islands. This over-interpretation contributed to a significant misdirection of the investigation, which in retrospect had wasted large amounts of police time and resources.

2. Ancestry analyses of the Minstead suspect's DNA

2.1. DNA data received before initiating USC SNP tests on the Minstead DNA

In November 2008, following discussions in Santiago with the SIO, USC received a Minstead suspect DNA sample from London's Metropolitan (Met) Police laboratory (UK Forensic Science Service, FSS). USC was also given DNA data from conventional FSS analyses plus two reports from a population genetics expert, dated 2004 and 2005, analysing DNAprint's 171-SNP genotype data. The 2005 report included a small population survey genotyping DNAprint SNPs in African Caribbean volunteers with grandparental or parental origins from five Caribbean regions. At no time were SNP identifiers, the 171-SNP profile, or reference population data provided to us by DNAprint.

2.1.1. Review and interpretation of conventional DNA analyses made by the UK FSS

The FSS provided an SGM + STR profile, Applied Biosystems Y-Filer™ profile (11/16 Y-STR genotypes) and mtDNA HVI/HVII sequence data. The Y-STR profile indicated an unequivocal R1b haplotype with a well-established West European geographic distribution. The mtDNA substitutions identified were HVS-I: 16189, 16192, 16223, 16278, 16294, 16309, 16390; and HVS-II 73, 143, 146, 152, 195, 263, 309 + C, 315 + C (unreadable sequence after 309 + C). This mtDNA profile was inferred to be haplogroup L2a1, with 2008 mtDNA population databases generally limited to HVS-I. Accounting for this limitation, we stated that L2a1 is a typical sub-Saharan African haplogroup common in Africa and the Americas, referencing our previous population analyses [1]. We reported mtDNA database matches in North Cameroon, Sao Tomé & Principe, Mauritania, African American samples, and the Caribbean. Therefore, uniparental variation in the Minstead DNA provided clear evidence of African maternal and European paternal co-ancestry. Considered in isolation, these data indicate an admixed background but cannot show whether African or European co-ancestry is the dominant component. Furthermore, a 2007 study of male surname groups from Yorkshire, UK, had observed an A1a African Y haplogroup in individuals who, in all other respects, were European [2], highlighting the Y chromosome is a single marker which can be atypical compared to overall genomic ancestry.

In Santiago, we were starting to develop statistical analyses of autosomal STR genotypes to potentially aid cold case investigations when, after routine profiling, further DNA extract is often unavailable. In such cases, a likelihood ratio (LR) using the cumulative random match probabilities (RMPs) for an autosomal STR profile can be informative. This approach was first suggested by the FSS in 2001 using six STRs [3], and we extended the Bayes LR analysis to include more STRs and a wider range of population comparisons [4]. Table 1 outlines the LR test of the suspect's SGM + profile (STR genotypes anonymised) applying allele frequency estimates from the CEPH human genome diversity panel (herein HGDP-CEPH). Note that STRs −2, −3, −4, −7 have higher likelihoods to be European than African, but STR−9 has a strongly indicative likelihood to be African. Despite this, the cumulative RMP indicated the Minstead profile was ~20 times more likely African than European, consistent with lower ancestry likelihoods from STRs compared to SNPs [4]. Nevertheless, as a preamble to evaluating the *DNAprint* SNP analyses, it was useful to have firm indications the suspect was African Caribbean or African American with a European patrilineage.

2.1.2. Review of population analyses of SNP genotypes from DNAprint

In 2004, *DNAprint* approached the Met Police offering their 171-SNP 'DNAWitness™' ancestry test. They obtained a complete SNP profile, and before independent analyses of this SNP data by a population genetics expert, *DNAprint* applied their own population analysis system using a tetrahedral plot of maximum likelihood estimation [5]. Analyses estimated co-ancestry proportions of 82% African; 6% European; 12% Native American; 0% East Asian. USC considered these values to be broadly in line with studies of African Caribbean populations [6,7], although the American co-ancestry component was much higher than observed in these studies, and the European co-ancestry lower.

Without data on *DNAprint*'s SNPs, we screened the scientific literature for information on the DNAWitness™ test to gain insights into the test's performance and component SNPs. Significantly, the most enlightening publication found was a 2007 commentary in Science by Bolnick et al. [8], which critically evaluated the accuracy of commercial direct-to-consumer (DTC) ancestry tests including *DNAprint's*, and outlined the limitations of the assumptions such tests adopt when predicting ancestry. Bolnick referenced the *DNAprint* test directly when describing the most paradoxical results reported to DTC customers: that 'most people from the Middle East, India and the Mediterranean region of Europe have significant proportions of Native American co-ancestry,

Table 1
Anonymized SGM+ STR profile data for the Minstead DNA reported by the UK FSS and cumulative random match probability analysis comparing African and European ancestries (HGDP-CEPH sample panel allele frequency estimates).

	African frequencies		European frequencies		Random match probabilities			
	Allele−1	Allele−2	Allele−1	Allele−2	Likelihood AFR	Likelihood EUR		
STR−1	0.332	0.356	0.300	0.259	0.2364	0.1554		
STR−2	0.302	0.129	0.229	0.262	0.0779	0.1200		
STR−3	0.207	0.086	0.154	0.147	0.0356	0.0453		
STR−4	0.124	0.020	0.063	0.086	0.0050	0.0108		
STR−5	0.342	0.084	0.228	0.040	0.0575	0.0182		
STR−6	0.173	0.139	0.155	0.073	0.0481	0.0226		
STR−7	0.051	0.232	0.103	0.130	0.0237	0.0268		
STR−8	0.069	0.193	0.018	0.321	0.0266	0.0116		
STR−9	0.347	0.347	0.249	0.120	0.2408	0.0598		
STR−10	0.190	0.190	0.187	0.187	0.0361	0.0350	**LR**	
				Cumulative values:	4.925E−14	2.442E−15	**20.16**	Times more likely African than European

when they do not' [8]. Bolnick concluded that the test likely considered some markers as diagnostic (i.e., specific to one region only), when, in fact, their ancestry information stems from differences in frequencies for SNP alleles shared between regions. Bolnick's commentary gave us the clearest indication that *DNAprint* SNPs over-estimated the Native American co-ancestry component in individuals who may or may not have such co-ancestry, and when they did, it was present at much lower proportions than the test inferred.

The independent population genetics expert provided a report on their analyses of the *DNAprint* SNP dataset in 2004 and a further report in 2005 included a specially commissioned small-scale Caribbean population survey genotyping the same SNPs. Both independent analyses performed a similar Bayesian approach to that of *DNAprint's* MLE test using the ADMIXMAP algorithm [9–11]. ADMIXMAP uses a Markov Chain Monte Carlo simulation (MCMC) to produce the posterior distribution of population metrics modelled from the data, including admixture co-ancestry proportions in the queried sample. Therefore, ADMIXMAP follows a near identical strategy to the widely used STRUCTURE algorithm [12,13]. The expert used ADMIXMAP to model co-ancestry proportions of each parent separately. However, in our view, a dataset comprising only 171 autosomal SNPs would not support haplotype analysis, as it does not allow closely sited SNP alleles to be reliably phased into single sequence strands. Notably, more recent analysis of an individual's parental ancestry requires vastly higher SNP densities (several hundred thousand loci) reliably phased to achieve 'chromosome painting' using haplotype reconstruction [14].

The 2004 report outlined a modelled 4-way admixture, estimating African-European-American-East Asian co-ancestry proportions and 3-way admixture, excluding East Asian co-ancestry. Results of these analyses are shown in Supplementary Tables S1A and S1B, respectively. The 4-way admixture model gave 64% African and 31% American proportions with marginal 3% European and 2% East Asian proportions, indicating a lack of compelling evidence for any European or East Asian co-ancestry in the Minstead DNA. Co-ancestry proportions for 3-way admixture were a near-identical 64% African, 33% American, 3% European. As with *DNAprint's* analyses, American co-ancestry proportions are much higher than were found in African Caribbean population surveys at the time [6,7]. A detailed study of Barbados, US Virgin Islands and Jamaica indicated mean co-ancestry proportions of 89.6%, 86.8%, 84.4% African; 10.2%, 10.6%, 12.4% European; 0.2%, 2.6%, 3.2% American, respectively [7]. Overall, studies give a consistent ~10% European co-ancestry proportion in Caribbean populations and a minimal Native American co-ancestry component. Additionally, we considered the African and American co-ancestry proportions from ADMIXMAP aligned more closely with individuals from continental South America than the Caribbean.

Supplementary Table S1B outlines co-ancestry proportions modelled for each parent, with the 'more African' parent having at least two-thirds African co-ancestry, and the 'less African' parent equal proportions of African and American co-ancestry, with at least 16% American proportions. The 2004 report concluded that the parent with equal African and American co-ancestry had possible origins in Guyana, St. Vincent, Trinidad or Belize, and such high American co-ancestry proportions are not observed in Jamaica.

USC viewpoints of this report were: i. an observed strong emphasis on American co-ancestry implying a highly informative subset of *DNAprint* SNPs signalled this ancestry, despite contrary indications in independent evaluations [8]; ii. novel use of estimated relative co-ancestry proportions to individualise each parent's background, although we did not understand how autosomal SNP data could not equally show the same co-ancestry proportions in both parents; iii. a level of specificity about where in the Caribbean the Minstead suspect's parents originated that was unmerited, given patterns of co-ancestry across the region were known to be relatively uniform. It is noteworthy that the report highlighted American co-ancestry proportions are high in Trinidad, one of the suggested places of origin, but higher minor European co-ancestry was independently observed here [6,7] and went undetected by ADMIXMAP - representing a conflicting pattern. In summary, we felt the data had been overfitted to attempt to provide quite specific ancestry information to the investigators centred on the suspect's parentage, which risked reaching erroneous conclusions, akin to the incorrect inference from *DNAprint* of American co-ancestry in East European, Middle East and South Asian individuals [8].

The 2005 report repeated ADMIXMAP analyses incorporating FSS uniparental data on the suspect's European patrilineage and African matrilineage. Using this knowledge, eight pairs of parental admixture ratios were modelled, as shown in Supplementary Table S2. The top three models all had very similar probabilities, each with 3-way co-ancestry in at least one parent and continuing to infer 10%−20% American co-ancestry. The report's second part gave, in our view, the most contentious population analyses, and we learnt from the SIO that it had given the Met Police what was perceived to be the 'breakthrough' investigative lead in efforts to identify the suspect [5]. A survey was commissioned of 104 Met Police volunteers with full grandparental origins from one of five Caribbean regions, combined with 36 samples with parental origins from these regions. The *DNAprint* SNP data from this survey was used to infer the most likely part of the Caribbean from which the suspect originated, as outlined in Fig. 1. ADMIXMAP produced -log likelihoods for an origin in each region which inferred a Trinidadian origin was seven times more likely than a Jamaican one. Our critique of this study highlighted several points: i. American co-ancestry proportions were much higher than surveys of Caribbean populations suggested, again highlighting a perceived lack of *DNAprint* specificity for American ancestry - the major factor distinguishing each region-of-origin; ii. using average co-ancestry estimates matched to individual co-ancestry proportions is imprecise, as the range of estimates

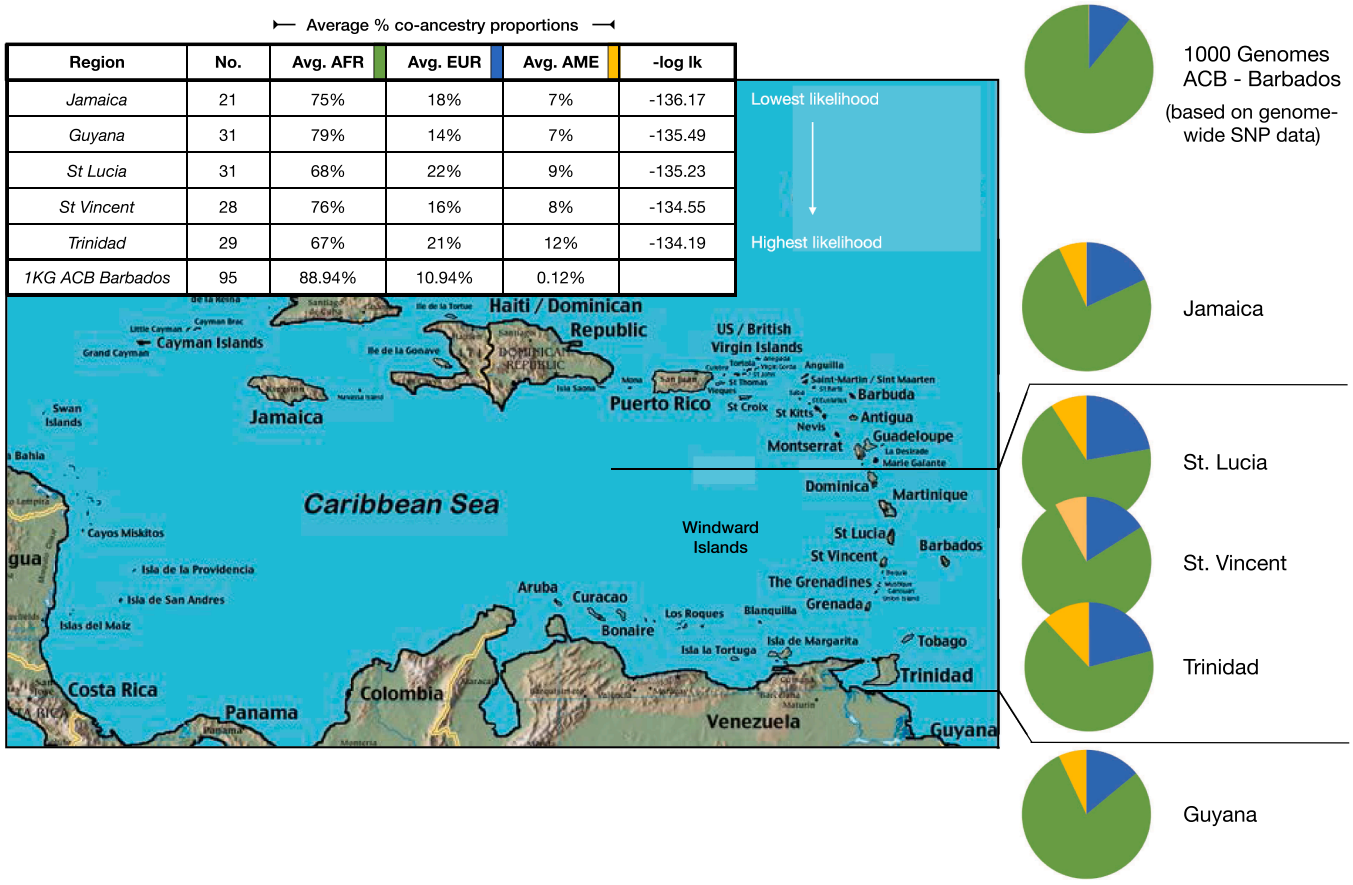


Fig. 1. Average 3-way co-ancestry proportions, estimated from ADMIXMAP analysis of the *DNAprint* 171-SNP data from a population survey of five regions of the Caribbean, three grouped in the Windward Island archipelago. Pie chart summaries are arranged geographically, apart from an independent 1000 Genomes (1KG) analysis of African Caribbeans from Barbados (ACB) based on genome-wide SNP data. Data in the table shows -log likelihoods (lk) of the Minstead suspect's parental origins from each location, ordered by ascending likelihood, i.e., origins in Trinidad are 7x more likely than origins in Jamaica (-log lk 134.19/-log lk 136.17). AFR: African, EUR: European, AME: Native American.

in any one region overlaps with the others; iii. no classification error was estimated using, e.g., cross-validation. Fig. 1 includes 'benchmark' Caribbean average co-ancestry estimates of Barbados (ACB) samples from 1000 Genomes' whole-genome analyses (millions of SNPs) representing a definitive point-of-reference for a typical Caribbean population [15], indicating almost no discernible American co-ancestry.

In discussions with the SIO, it was clear that the Minstead investigation had been misdirected by the above reports in the four years prior to our evaluations. In particular, the reports' focus on parental co-ancestry patterns and inference of the suspect's Windward Island origins, based on minimal likelihood differences and SNPs with contested Native American specificity, had concentrated police resources firmly in a particular direction. This misdirection meant eight Met Police officers travelled to Trinidad, Barbados, Tobago, St. Lucia and St. Vincent, to gain information that might explain extensive gaps in Minstead criminal activity. We advised to evaluate in detail the co-ancestry estimates previously made, by genotyping USC ancestry SNPs. Our findings could redirect the investigation framework away from trying to identify a very specific region-of-origin towards a broader inference of the suspect's likely background. At the time our SNP analyses started, we had developed and optimised a 29-plex SNaPshot assay analysing Native American population variation which included SNPs with extremely differentiated allele frequencies between American and other populations. Therefore, we were confident more detail could be added to the co-ancestry patterns previously inferred using *DNAprint* SNPs, and we could gauge the accuracy of those inferences.

2.2. USC autosomal SNP tests analysing co-ancestry patterns in the Minstead suspect

2.2.1. Bayes analysis of USC ancestry-informative SNPs

The Minstead DNA sample was first genotyped with SGM + to check data concordance and ensure no evidence of mixed DNA was detected. USC planned four SNaPshot-based SNP assays: an established 34-plex ancestry test [16]; a panel of SNPs to detect and characterise Native American ancestry named PIMA; and two pigmentation-informative FDP tests named SHEP−1 and SHEP−2. PIMA was designed to complement the 34-plex SNPs differentiating African, European and East Asian ancestry, by adding 29 SNPs with strongly contrasting allele frequencies between American and these populations. The 2008 PIMA panel was slightly different from the final multiplex published in 2020 [17], and Supplementary File S2 gives population and marker details of the 29 PIMA SNPs genotyped here. PIMA included one X chromosome SNP with a fixed G allele frequency in Africans (i.e., close to 1.0 in Africans and 0.0 in all other populations), so potentially adding matrilineal data. Finally, both SHEP assays included several ancestry markers plus FDP SNPs with highly contrasted allele frequencies between Europeans and other populations, so were included in the ancestry analyses.

A total of 27 of the 34-plex SNPs and 14 PIMA SNPs, including the X-SNP, were successfully genotyped. A subset of 12 ancestry-informative SHEP SNPs were added, compiling 53 SNPs to evaluate ancestry. Visual scrutiny of genotypes failed to identify any American-informative alleles, so SNP-by-SNP statistical analyses were made of the likelihood of each SNP's genotype being observed in an individual with either of

three possible ancestries, e.g., in PIMA ASNP1, the likelihood of the CC genotype detected in the Minstead DNA, given an American (AME), European (EUR) or African (AFR) ancestry, was evaluated using AFR/AME, AFR/EUR, EUR/AME LR calculations. The CC genotype HGDP-CEPH frequency estimates are AFR 0.970, EUR 0.097, AME 0.0001 (a 'seeded' value, as HGDP-CEPH Americans have zero C allele frequency), producing an AFR/AME LR of ~9.7 billion times more likely African than American, and a EUR/AME LR of 969 million times more likely European than American. Individual SNP analyses helped keep focus on each marker showing extreme allele frequency differences, lost in the cumulative LR likelihood and single ancestry inference made by Snipper [16].

Table 2 lists the 16 most informative autosomal SNP genotypes obtained, ranked in descending order of LR values for one or two most likely ancestries in each SNP. The other 47 ancestry-informative SNPs successfully genotyped gave lower LR values and are not detailed. None of the genotypes in Table 2 provided any statistical support for American ancestry with all ten PIMA SNPs showing an absence of American-indicative alleles. SHEP PSNP21 gave the strongest support for European co-ancestry with the detected T allele almost reaching population specificity, but present in 0.5% of HGDP-CEPH African genotypes. Overall, 9 of the 16 Table 2 SNPs gave LR values higher than '100 times more likely' in both African and European ancestries against American ancestry.

Therefore, USC Bayes analyses strongly indicated a dominant African co-ancestry component and indications of a minor European co-ancestry component. There was no statistical evidence for American co-ancestry in the Minstead suspect, with the highest LR values for ASNP2-G and ASNP12-C alleles having equivalent likelihoods for either African or American ancestry. Lastly, the PIMA X-SNP G allele detected gave an LR of 100 times more likely African (0.96 HGDP-CEPH allele frequency) than European (>0.01), and 80 times more likely African than American (0.012), indicating an African X chromosome in the Minstead suspect, and aligning with the mtDNA L2a1 matrilineal haplogroup.

2.2.2. *Genetic cluster analysis of USC ancestry-informative SNPs using STRUCTURE*

In 2008, we already routinely applied genetic cluster analyses with the STRUCTURE algorithm [12,13], and it provided reliable indications of minor co-ancestry when present at levels above 10% in the analysed individual. To gauge this reliability, we previously used STRUCTURE to explore HapMap admixed ASW and MEX samples (African Americans from SW USA, individuals with Mexican ancestry resident in Los Angeles, respectively) using 34-plex, PIMA, or larger SNP panels and in-silico genotype data from HapMap and the HGDP-CEPH panel [18]. STRUCTURE uses an identical MCMC simulation approach to ADMIXMAP to produce a posterior distribution of metrics modelled from the data, including the optimum number of genetic clusters for the prescribed range (the K parameter).

STRUCTURE analysis was run on the 53-SNP dataset with parameter settings we had standardised for forensic SNP panel analyses (K values 2–4, analysing the POPFLAG=0 Minstead DNA plus three POPFLAG=1 reference datasets of HGDP-CEPH Africans, N = 102, Europeans, N = 158, and Native Americans, N = 64; 200,000 burn-ins; 200,000 MCMC steps; 5 reiterations per K value of the admixture and independent allele frequency model). The optimum K value of 3 from Evanno's

Table 2
Individual SNP genotype likelihood ratios (LRs) for AFR/EUR (African/European) AFR/AME (/American) and EUR/AME comparisons in 15 anonymised ancestry-informative ASNPs and one phenotyping PSNP (with a highly contrasting European allele frequency). Strongly indicative LR values in bold, reciprocal LRs (i.e., reverse ratios) in gray boxes. The only high LR value for Native American vs European ancestry was identical to that for African vs European in ASNP2. Ref. reference allele, Alt. alternative. LR values less than 3 not shown. Zero allele frequencies marked * were set to 0.00001 to obtain a non-zero LR.

Assay	SNP ID	Chr.	American-informative allele	Minstead genotype	SNP alleles		African allele frequency	European allele frequency	American allele frequency	African genotype frequency	European genotype frequency	American genotype frequency	LR for AFR/EUR genotype	LR for AFR/AME genotype	LR for EUR/AME genotype	Verbal description of individual SNP genotype likelihood ratio tests
PIMA	ASNP1	1	T	C/C	Ref:	T	0.015	0.689	1	0.97	0.097	0	10	**9.7 billion**	**969,076,900**	C/C genotype 9.7 billion x more likely in an individual with AFR ancestry than AME (969 million x more likely EUR than AME)
					Alt:	C	0.985	0.311	0.000*							
34-plex	ASNP2	15	G	G/G	Ref:	A	0.046	0.997	0.04	0.911	0	0.922	**83,661**		0.00001	Similar odds for G/G genotype >80,000 x more likely in an individual with AFR ancestry than EUR, AME ancestry than EUR
					Alt:	G	0.955	0.003	0.96						84,628	
34-plex	ASNP3	1	T	C/C	Ref:	T	0.071	0.997	0.98	0.864	0	0	**79,302**	2,159	0.03	C/C genotype >79,000 x more likely in an individual with AFR ancestry than EUR (>2,000 x more likely AFR than AME)
					Alt:	C	0.929	0.003	0.02						36.73	
34-plex	ASNP4	10	G	G/C	Ref:	G	0.692	1	1	0.426	0	0	**21,317**	**21,317**		G/C genotype >21,000 x more likely in an individual with AFR ancestry than AME or EUR
					Alt:	C	0.308	0.000*	0.000*							
SHEP 1	PSNP21	10	-	G/T	Ref:	G	0.995	0.871	1	0.005	0.112	0	0.04	497	**11,243**	G/T genotype >11,200 x more likely in an individual with EUR ancestry than AME
					Alt:	T	0.005	0.129	0.000*				22.6			
PIMA	ASNP5	16	A	G/G	Ref:	A	0.121	0.487	0.99	0.772	0.263	0		**7,723**	**2,635**	GG genotype >7,700 x more likely in an in/dividual with AFR ancestry than AME (>2,600 x more likely EUR than AME)
					Alt:	G	0.879	0.513	0.01							
PIMA	ASNP6	22	A	G/G	Ref:	G	0.803	0.748	0.02	0.645	0.56	0		**1,612**	**1,400**	Similar odds for G/G genotype >1,400 x more likely in an individual with AFR ancestry than AME, and EUR ancestry than AME
					Alt:	A	0.197	0.252	0.98							
PIMA	ASNP7	5	T	C/C	Ref:	T	0.015	0.434	0.97	0.97	0.321	0.001		**1,078**	356	C/C genotype >1,000 x more likely in an individual with AFR ancestry than AME (>350 x more likely EUR ancestry than AME)
					Alt:	C	0.985	0.566	0.03							
PIMA	ASNP8	11	G	A/A	Ref:	A	0.985	0.533	0.04	0.97	0.284	0.002	3.41	**606**	177	A/A genotype >600 x more likely in an individual with AFR ancestry than AME
					Alt:	G	0.015	0.467	0.96							
PIMA	ASNP9	2	A	G/G	Ref:	G	1	0.927	0.05	1	0.86	0.003		**400**	**343**	Similar odds for G/G genotype >340 x more likely in an individual with AFR ancestry than AME, and EUR ancestry than AME
					Alt:	A	0	0.073	0.95							
PIMA	ASNP10	17	A	G/G	Ref:	G	0.975	0.805	0.07	0.95	0.647	0.005		**194**	**132**	Similar odds for G/G genotype >130 x more likely in an individual with AFR ancestry than AME, and EUR ancestry than AME
					Alt:	A	0.025	0.195	0.93							
PIMA	ASNP11	7	C	T/T	Ref:	C	0.111	0.182	0.93	0.79	0.669	0.005		**161**	**136**	Similar odds for T/T genotype >130 x more likely in an individual with AFR ancestry than AME, and EUR ancestry than AME
					Alt:	T	0.889	0.818	0.07							
34-plex	ASNP12	13	C	T/T	Ref:	T	0.934	0.967	0.12	0.873	0.935	0.014		**60**	**64**	Similar odds for T/T genotype >60 x more likely in an individual with AFR ancestry than AME, and EUR ancestry than AME
					Alt:	C	0.066	0.033	0.88							
34-plex	ASNP13	5	C	C/C	Ref:	C	1	0.136	0.94	1	0.018	0.884	**54**		0.02	Similar odds for C/C genotype 54 x more likely in an individual with AFR ancestry than EUR, 47 x more likely AME than EUR
					Alt:	G	0	0.864	0.06						47	
PIMA	ASNP14	8	T	C/C	Ref:	C	1	0.924	0.16	1	0.854	0.026		39	33	Similar odds for C/C genotype 39 x more likely in an individual with AFR ancestry than AME, 33 x more likely EUR than AME
					Alt:	T	0	0.076	0.84							
PIMA	ASNP15	12	T	C/C	Ref:	C	1	0.924	0.22	1	0.853	0.048		20	17	Similar odds for C/C genotype 20 x more likely in an individual with AFR ancestry than AME, 17 x more likely EUR than AME
					Alt:	G	0.298	0.497	0.99							

'likelihood of K' calculation [19], gave average cluster membership proportions of: African cluster 0.847; European 0.12; American 0.033. The cluster and triangular plots obtained from the Minstead DNA (rightmost column and yellow point, respectively) are shown in Fig. 2. These were interpreted to indicate major African co-ancestry of ~85–88% and minor European co-ancestry of ~12–15%, with the American co-ancestry component too small to accurately measure. Therefore, STRUCTURE cluster membership proportions contradicted co-ancestry proportions estimated from the *DNAprint* SNPs but were in line with values estimated from studies of Barbados, US Virgin Islands and Jamaica [7].

2.3. Summary ancestry findings from USC reported to the Minstead investigators

We stated that no USC SNP alleles indicating Native American origins (i.e., with extreme allele frequency differences between American and other populations) were detected and the ~3% American cluster membership proportion estimated by STRUCTURE further underlined an absence of American co-ancestry in the Minstead suspect. Therefore, the estimated American co-ancestry proportion was found to be substantially less than the 33% suggested by the expert's reports, or absent altogether. The FSS Y-STR profile and small proportion of European-indicative SNP alleles detected suggest a minor European ancestry component. The 85%:12–15% African:European co-ancestry proportions estimated using STRUCTURE were in line with very similar estimates from independent Caribbean population studies [7]. These proportions are inconsistent with the Minstead suspect being sub-Saharan African, and consistent with having African Caribbean origins, although a Continental American origin could not be ruled out.

3. USC SNP tests to explore patterns of pigmentation traits in the Minstead suspect

3.1. Analysis of SHEP SNPs associated with eye, hair and skin colour variation in Europeans

During discussions with the SIO, it was agreed SNP tests providing data on pigmentation traits could be potentially informative, given USC had detected a proportion of European co-ancestry. Unlike African populations, Europeans show widespread variability in eye and hair colour. Skin pigmentation has less variability, although paler skin in northwest Europeans is widely recorded and influences the expression of e.g., freckling and skin sensitivity to sun. We have always advocated against inferring the appearance of individuals from their ancestry alone, and the SIO stated some eyewitnesses implied the suspect had pale skin. Therefore, we agreed to apply tests genotyping SNPs most strongly associated with European eye, hair and skin pigmentation variation. By 2008, we finished developing two 'SHEP' eye and hair colour predictive SNP assays (published in 2013 [20]) but not a skin colour panel (published in 2014 [21]). We also lacked the Bayes predictive system later incorporated into Snipper (https://mathgene.usc.es/snipper4/SHE_marker_panels.php). A strategy was adopted comparing the SHEP genotypes detected with alleles most closely associated with European pigmentation variation in large-scale genome-wide association studies (GWAS) [22–24] and multiple additional studies on SNPs strongly associated with blue eye colour and light hair colours in Europeans [25–27]. SNP data from skin colour variation studies [28–30] was used to identify alleles associated with light skin in Europeans, already known to be different to equivalent loci in East Asians [29].

Table 3 lists 19 (of 35) anonymised SHEP SNP genotypes detected in the Minstead suspect with the highest association values to European pigmentation variation. SNPs comprised 14 from the landmark Sulem GWASs [22,23], one from Han [24], the blue eye colour studies by Sturm et al., Kayser et al. and Eiberg et al. [25–27] and three skin colour SNPs from Stokowski [30]. The 2/19 Minstead suspect alleles matching eye and hair colour associations identified in European studies are highlighted in grey, comprising PSNP5-C (homozygous) weakly associated with brown eyes and PSNP14-C (heterozygous) moderately associated with blue eyes/blond hair. Therefore, our report emphasised the absence of strongly blue eye-associated alleles (PSNP6-A, PSNP16-G) and light hair-associated alleles (PSNP1-T, PSNP2-C, PSNP4-A, PSNP6-A, PSNP12-T, PSNP13-A), stating that no SNP alleles, except PSNP14-C, were detected which would indicate non-African eye and hair pigmentation in the suspect.

Analysing skin pigmentation, the key Stokowski skin depigmentation loci (SNPs 17–19) all had homozygote non-European alleles, interpreted to indicate the suspect's European co-ancestry had not influenced his skin colour, with little indication it was significantly lighter than most African origin individuals.

3.2. Additional analysis of MC1R variation

Four of the five most commonly occurring *MC1R* SNPs associated with red hair were genotyped by the SHEP assays. Three, including PSNP8 and 9 in Table 3, were 'R' variants with dominant effect on the red hair colour and fair skin phenotype (RHC) as homozygotes or multiple SNP allele combinations (compound heterozygotes) [31]. A fourth SNP (PSNP21 in Table 2) was an 'r' variant, i.e., only affecting RHC expression when the SNP's T allele occurs as a compound heterozygote with another *MC1R* variant. The phenomenon of R/r compound heterozygosity leading to RHC was seen in the first forensic FDP test developed by Grimes et al. in 2001 [32]. Grimes' assay genotyped five 'r' SNPs (residue changes V60L, V92M, R155W, R163Q), and five 'R' SNPs

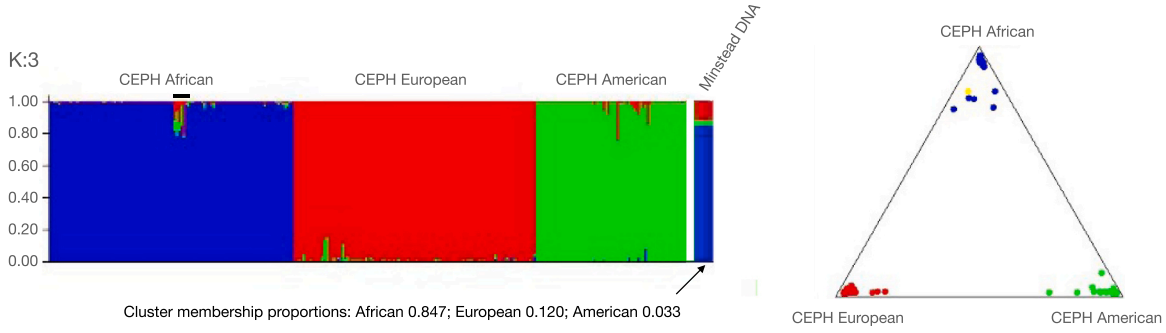


Fig. 2. Genetic cluster membership proportions from STRUCTURE analysis of the Minstead DNA using the genotypes of 53 ancestry-informative SNPs in USC tests and assigning K:3 clusters. The single Minstead cluster plot in the rightmost column, is widened to show proportions more clearly. The STRUCTURE triangular plot is shown right (each vertex represents cluster membership proportions of 1) with the Minstead DNA position marked by the yellow point. African population columns highlighted with a black bar are five HGDP-CEPH Khoisan samples included with the other African populations, despite showing atypical allele frequency distributions. The Khoisan samples are evident as the five blue outlier points in the triangular plot.

Table 3
Nineteen anonymised FDP SNPs that have the strongest associations with eye, hair and skin colour phenotypes in key genome-wide association studies. Listing the odds ratios, the associated allele and the Minstead DNA genotype detected by USC SHEP tests. Note only the first SNP allele of the TGT haplotype was typed, so no link was made between the Minstead genotype and associated phenotypes. No equivalent metrics to odds ratios were recorded in the later GWASs of Han, Stokowski, or the multiple eye colour associations studies of PSNP16.

GWAS	Anonymised SNP No.	Gene	Odds Ratio Sulem Eye	Odds Ratio Sulem Hair	Odds Ratio Sulem Skin	Associated allele	Minstead genotype	Pigmentation phenotype associations
Sulem 2007	PSNP1	*SLC24A4*	1.15	2.56	1.21	T	CC	T-allele associated with blue eye color / blond hair / sun sensitivity in Europeans
	PSNP2	*KITLG*	1.13	2.32	/ | °0.89	C	AA	C-allele associated with blue eye color / blond hair in Europeans
	PSNP3	*6p25.3*	1.11	0.69	1.21 | °1.40	A	CC	A-allele associated with sun sensitivity in Europeans
	PSNP4	*TYR*	1. 20	1.29	1.26 | °1.13	A	GG	A-allele associated with blue eye color / blond hair / sun sensitivity in Europeans
	PSNP5	*TYR*	/	0.85	/ | °1.32	C	CC	C-allele associated with brown eyes / °freckles in Europeans
	PSNP6	*OCA2*	29.43 | *6.74	4.94	1.24 | °1.16	A	CC	A-allele strongly associated with blue eyes (vs brown; *vs green), associated with light hair in Europeans
	PSNP7	*OCA2*	6.90	1.92	1.30 | °0.84	TGT	T only	TGT haplotype moderates HERC2 G-allele (SNP16) associated with blue eye color in Europeans. Also associated with blond hair / sun sensitivity
	PSNP8	*MC1R*	1.15	‘7.86	2.30 | °2.63	T	CC	T-allele strongly associated with red hair. Associated with sun sensitivity in Europeans
	PSNP9	*MC1R*	1.37	‘12.47	2.94 | °4.37	T	CC	T-allele strongly associated with red hair. Associated with sun sensitivity in Europeans
Sulem 2008	PSNP10/11	*ASIP*	-	1.76-1.46	1.76 | °1.95	G–T diplotype	G–G G–G	G–T AH diplotype associated with red-blond hair / sun sensitivity & freckles in Europeans
	PSNP12	*TPCN2*	-	2.49	-	T	AA	T-allele associated with blond hair in Europeans
	PSNP13	*TPCN2*	-	1.60	-	A	CC	A-allele associated with blond hair in Europeans
	PSNP14	*TYRP1*	1.40	1.29	-	C	CT	C-allele associated with blue eyes/blond hair in Europeans
Han	PSNP15	*IRF4*				T	CC	T-allele associated with blue eyes / brown hair / freckling & lighter skin in Europeans
Multiple	PSNP16	*HERC2*				G	AA	G-allele strongly associated with blue eye color in Europeans
Stokowski	PSNP17	*SLC45A2*				G	CC	G-allele strongly associated with light skin pigmentation in Europeans
	PSNP18	*SLC24A5*				A	GG	A-allele strongly associated with light skin pigmentation in Europeans
	PSNP19	*TYR*				A	CC	A-allele associated with light skin pigmentation in Europeans

- No association detected by the Sulem 2008 GWAS for that pigmentation phenotype / Balanced odds for each SNP allele.
° Odds ratio for freckling and/or sun sensitivity. * Odds ratio for blue vs green eyes. ‘ Odds ratio for red vs non-red hair.

(D84E, R142H, R151C, R160W, D294H), identified as principal SNP variants underlying the RHC phenotype (see Fig. 1 of [31]). Grimes observed 4/32 RHC samples were compound heterozygotes with ‘r’ variants, and our genotyping detected one RHC-associated T allele in ‘r’ SNP PSNP21, but the three ‘R’ *MC1R* SNPs in SHEP had non-RHC alleles, so we reported an absence of strong indications of red hair in the Minstead suspect.

The SIO related an eyewitness sighting of a red-haired individual near a Minstead crime-scene and asked to clarify if the PSNP4 T allele detected would affect RHC expression. This led to checks of published studies of African Caribbeans with the RHC phenotype, including sequencing studies of four red-haired Jamaicans by McKenzie et al. [33]. The subjects of McKenzie’s study all self-identified as African Caribbeans with auburn/reddish hair, ‘rust-coloured’ complexion and freckles (McKenzie’s descriptions). Sequencing detected r/R compound heterozygotes in each subject (residue changes G89R/R151C, codon23-STOP/R160W, codon195-frameshift/D294H, codon23-STOP/R151C), and McKenzie concluded this indicated substantial ‘sampling’ of European *MC1R* diversity amongst Jamaicans with RHC. McKenzie’s study subject description stated: ‘The phenotype is striking; it is very easy to distinguish these individuals from black persons of lighter or darker complexion because there is a reddish-orange/rust coloured hue to the skin’. Another more extensive population survey of *MC1R* variants by Gerstenblith et al. [34] detected SNPs creating five non-synonymous substitutions (S47I, L99I, R163Q, F196L, D294H) in 276 Africans (collective allele frequency 0.036). Subsequently, we decided the possibility to detect an atypical combination of pigmentation phenotypes in the Minstead suspect merited developing a sequence analysis regime *de-novo* for the whole *MC1R* gene to attempt location of possible additional variants not detected by SHEP assays.

To sequence *MC1R*, the single exon and portions of the 5’ and 3’ UTRs were targeted in a 1060 nucleotide (nt) sequence - *MC1R* being one of the smallest human genes. Three primer pairs were designed in-house to sequence 378 nt, 339 nt, 454 nt, overlapping fragments (see Supplementary File S3). Sequencing only succeeded for the third sequence span, which showed an absence of RHC-associated alleles in 8 recognised SNP sites. Given the *MC1R* SNPs detected by SHEP assays have a ~0.37 combined European frequency, and the two ‘R’ SNPs not detected only ~0.02, it was highly improbable on allele frequency grounds alone, that another *MC1R* variant in a compound heterozygote was present from the estimated 12% European co-ancestry.

3.3. Final genetic analyses of a possible pale skin African phenotype

Discussions with the SIO about the lack of any evidence for pale skin from RHC-linked variants, led to the decision to analyse remaining DNA extract for oculocutaneous albinism - a rare trait in African populations (frequency: 1 in 4000 in sub-Saharan Africans and African Americans), but with a distinct phenotype similar to McKenzie’s description [33]. Two forms of oculocutaneous albinism in Africans are termed OCA1 (*TYR* mutations, 1 in 40,000 frequency) and OCA2 (*OCA2* mutations or deletions, 1 in 4000). A founder 2.7-kb interstitial deletion in exon−7 of the *OCA2* gene is the one common variant responsible for ~50% of OCA2 phenotypes [35]. We opted to use the 4-primer system developed by Manga et al. [36] to detect the *OCA2* exon−7 deletion. The PCR test outlined in Supplementary File S4, used a common reverse primer (MHB71) linked to one forward primer (MHB72) annealing inside the exon−7 sequence creating a 240 nt fragment when intact, and two forward primers (MHB51, MHB107) bridging the deleted sequence creating 820 nt and 420 nt fragments. The test produced a single 240 nt fragment, giving a clear indication of homozygous wild-type (intact) *OCA2* exon−7 sequences in the Minstead suspect.

3.4. Summary of pigmentation phenotype findings from USC reported to Minstead investigators

In the USC report, the homozygous genotypes of PSNPs 6 and 16 for eye colour, and PSNPs 17 and 18 for skin colour, were described as

significant findings. It was concluded there was no evidence the Minstead suspect had blue eyes. In the absence of any indications of light hair colour-associated alleles detected in European GWAS subjects with these phenotypes, there was no evidence for non-African hair colour. We reported that PSNP17-G and PSNP18-A alleles were not detected, so the Minstead suspect had no indications of lighter skin tone compared to most African Caribbean or African American individuals, notably those with higher proportions of European co-ancestry (up to ~30% in both populations [15]). Sequence analysis of *MC1R* was unsuccessful for 2/3rds of the coding region, but with a combined frequency for undetected variants of only ~2.5% in Europeans, it was highly unlikely the PSNP21-T allele formed a compound heterozygote with another undetected variant. The *OCA2* exon−7 deletion underlying the rare African OCA2 phenotype was not detected.

4. Ethical and societal aspects of investigative practice in Operation Minstead

Using forensic DNA for ancestry analysis (in the context of population admixture, this paper deliberately favoured the term 'analysis' rather than 'inference' or 'prediction') inevitably raises ethical concerns about how a balance can be struck between intelligence-led investigation, that focusses on particular ancestry groups, and racialised suspicion. Suggesting to the police a likely ancestral background for a suspect can risk stigmatisation of members of the community with the same population origins, who may feel each one of them can be viewed as a suspect. FDP analysis to derive a set of suggested phenotypes is less prone to accusations that its use can lead to stigmatisation but has ethical concerns around privacy issues and the extent to which FDP SNP tests can obtain knowledge about individual risk factors such data may coincidentally reveal (e.g., certain *MC1R* variants have been associated with heightened risk of melanoma).

Supplementary File S5 discusses four ethical aspects of Operation Minstead, focussing on: coercive tactics used during a DNA mass screen; handling risks of racial stigmatisation; operational proportionality – the use of limited resources to efficiently investigate major crime; and the moral issue of obstructing cooperation and data exchange amongst forensic experts. Discussions of the ethical aspects of SNP-based phenotyping of physical characteristics have been widely published ([37, 38] and additional FDP discussion and review papers are listed in Supplementary File S5.

5. Discussion – lessons learnt since Minstead

Choosing to outline a historical criminal case that culminated in the conviction of Grant in 2011, but where DNA analyses did not meaningfully progress the investigation, may appear strange, particularly as the USC Forensic Genetics Unit has since successfully resolved many cold cases and active investigations by applying the type of ancestry and phenotyping SNP tests described here [39]. We decided our involvement in Operation Minstead had provided insights for applying forensic SNP tests, which have helped to shape how we approach requests for such tests in criminal investigations since 2008. In the absence of paragraph headings, the following five lessons learnt from Minstead are discussed:

- Using averaged co-ancestry proportions to assign an individual to one of several admixed populations is inappropriate when most individuals across the populations compared have overlapping co-ancestry estimates. For this reason, accurate co-ancestry analysis depends on the most extensive possible sampling of contributing populations, rather than admixed populations themselves.
- When alternative expert advice is given to investigators the sharing of all genetic data is essential for proper scrutiny of the inferences each party makes
- Investigators hope for uncommon characteristics in the suspect and complete MC1R SNP analysis offers the best opportunity to detect highly distinctive RHC phenotypes (which could have proved decisive for Minstead)
- Enhancements to Snipper and addition of extra functionality, including STR-based ancestry inference, continue to progress with emphasis on flexible data input allowing users to adapt markers and populations to casework requirements
- Ancestry inference of admixed individuals and analysis of co-ancestry require the most strongly differentiated allele frequencies possible, given the constraints on forensic SNP multiplex sizes

Following previous experience with the major forensic investigation of the 2004 M–11 Madrid bomb attacks [40], we have placed particular emphasis on the need for caution when likelihoods from SNP-based ancestry analyses are low or close to values where the risk of predictive error becomes substantial. Therefore, we were circumspect in our interpretation and view of the geographic precision suggested by the independent expert for Grant's origins in the southern Caribbean Windward Islands archipelago. To illustrate the risk of error when attempting to predict a person's origin from a specific region, a review of co-ancestry estimates of 1000 Genomes' Admixed American populations [15] is informative. Genetic cluster analyses of African Caribbeans from Barbados (ACB) and African Americans from SW USA (ASW) (Supplementary File S6A, individual cluster membership data in Fig. 2a of [15] from Adam Auton, 1000 Genomes Project) show different average African and European co-ancestry proportions, comprising: ACB, AFR 0.89, EUR 0.11 vs ASW, AFR 0.78, EUR 0.19. However, given standard deviations of: ACB, AFR $+/-0.07$, EUR, $+/-0.07$ vs ASW, AFR $+/-0.12$, EUR $+/-0.09$; most individual co-ancestry proportions in each population have overlapping value ranges and will not reliably infer any one person's specific origins in the Caribbean or US mainland. Supplementary File S6B shows 1000 Genomes' genome-wide analysis of ACB and ASW found the minimum detectable proportion of American co-ancestry (<0.0001) in 32% of ASW and 68% of ACB. Finally, Supplementary File S6C illustrates that the other four admixed American populations of 1000 Genomes (see Fig. 2 of [41]) share very similar admixture profiles, dominated by European and American co-ancestry. Supplementary File S6C shows a large proportion of individuals are indistinguishable when comparing populations with very similar admixture profiles. Since reviewing the Minstead analyses, we have strongly advocated against using averaged co-ancestry proportions to differentiate admixed populations that show a wide range of mainly overlapping values. We believe a significant factor hindering the investigation was the failure to consider Jamaica alongside the Windward Islands as Grant's possible region of origin. Once arrested and identified, it became known Grant was born in Jamaica in 1957 and moved to London with his father when fifteen and then began his criminal career. We have seen overinterpretation of population data to give an overly precise prediction of ancestral origin from other laboratories on several occasions, but in Minstead the information was key to directing the investigation to very specific regional enquiries which wasted police time, and well before our independent scrutiny. Following our original SNP analyses based solely on HGDP-CEPH data, availability of good quality reference population databases for forensic ancestry analysis has improved both in number and geographic scope. However, it is noteworthy that we must still rely on 64 Native American and 28 Oceanian samples from HGDP-CEPH to complete continental STRUCTURE-based analyses, and although the six admixed American populations in 1000 Genomes (see Supplementary File S6) are useful for evaluating patterns of admixture in the Americas, significant gaps remain in wider population coverage in the above regions (notably, North America and Polynesia), as well as the more populous areas of East Africa, Middle East and Island SE Asia. Finally, mention should be made of the potential for ambiguous inferences when identifying African-European co-ancestry, since this was a regular pattern observed in North African individuals until dedicated Middle East-informative SNP panels [42] were developed ten years after Minstead.

The lack of identifying information on *DNAprint's* SNPs, reference population data and the Minstead DNA SNP profile, was a major

hinderance to the advice we could give the SIO on how to handle the population genetics data provided in 2004. We suspected *DNAprint* SNPs overestimated American co-ancestry in Caribbean populations and the Minstead suspect, so when Bolnick [8] highlighted the numerous incorrect inferences from their DTC ancestry tests we were confident in applying PIMA SNPs to confirm these inaccuracies. Interestingly, when the *DNAprint* SNPs were published [43] only three SNPs (rs2814778, rs2065160, rs730570) overlapped with USC marker sets. Except for rs2814778, we did not identify other *DNAprint* SNPs with extreme frequency differences able to produce LR values as high as those of Table 2. The *DNAprint* assays used the *SNPstream* genotyping system [44], which has not been validated for forensic use by evaluating sensitivity, reproducibility, genotyping concordance, etc. Fortunately, since the *DNAprint* 'black box' we have not encountered another case where details of a DNA test were withheld to protect commercial interests. Nevertheless, proprietary genomic analyses used by some private investigative genetic genealogy (IGG) services restrict independent scrutiny of these novel DNA analysis techniques, essential for their forensic validation.

In discussions with the SIO, it was important to emphasise that the Minstead suspect had no ancestry or phenotype features that would differentiate him from the approximately 340,000 African Caribbeans living in the Greater London area (the 2001 UK census recorded 343,567 who self-identified as "Black Caribbean"). Although this amounted to about 5% of London's populace at that time and therefore substantially reduced the investigation's male suspect pool, amongst the African Caribbean community Grant might be described as 'nondescript', i.e., lacking distinct or distinguishing externally visible characteristics. Furthermore, Jamaicans make up by far the largest proportion of London Caribbeans, and it was not clear to us the extent to which suspects with Jamaican origins had been given lower priority in the investigation. We were prepared to exhaust all possible skin colour and RHC analysis approaches to try to reconstruct Grant's likely appearance. However, this was based on the available knowledge at the time on pigmentation expression in Europeans and little is still known about how European SNP variation is affected in a predominantly African genomic background. Therefore, our interpretation of SHEP SNP genotypes focussed on the near-complete absence of alleles known to be closely associated with European pigmentation patterns from major GWASs. The pursuit of full coverage of the~100 recorded *MC1R* coding SNPs [32], although ultimately unsuccessful, was considered worthwhile, since African Caribbeans with *MC1R* variants as compound heterozygotes show distinctive hair and skin colour phenotypes [33]. We believe *MC1R* sequencing should be incorporated into FDP tests and note the comprehensive 2025 review of *MC1R* by Branicki et al., shows sequence analysis with Oxford Nanopore Sequencing is cost-effective and straightforward [45]. More importantly, our experience of several cases since Minstead demonstrates detection of red hair in a suspect is amongst the most useful indications of a distinct phenotype that can be given to investigators.

Influenced by our experiences with Minstead, Snipper has since evolved by offering the more flexible option to analyse 'freeform' SNP combinations (e.g., combined PIMA−34-plex-SHEP subsets), not just fixed forensic panels [46]. A profile upload comprising the user's own choice of SNPs and reference populations, produces a Bayes LR value linked to Principal Component Analysis (PCA). Both have limited power to detect admixture, with the LR value simply representing the highest likelihood for the dominant co-ancestry component, and PCA plots limited to signalling only parental co-ancestry when intermediate positions are observed. At USC, STRUCTURE remains the mainstay for co-ancestry estimation and Snipper now offers direct preparation of formatted STRUCTURE data input from the original profile and reference population uploads - making the process fast, straightforward and error-free. Lastly, the advantages offered by pre-evaluation of the Minstead suspect's ancestry from the FSS autosomal STR profile, prompted us to develop a formal Bayesian likelihood calculator for Snipper (https://mathgene.usc.es/snipper4/frequencies_new.php); analysing up to 32 STRs with allele data for seven reference populations from the dedicated HGDP-CEPH database in SPSmart's pop.STR (http://spsmart.cesga.es/popstr.php).

The estimation of co-ancestry proportions in a person from an admixed background was clearly a dominant feature of each independently made population analysis. Since 2008, we have continued to improve the SNPs chosen for forensic ancestry analysis by prioritising markers with allele frequency extremes (broadly defined as $\geq$0.95 in one population, $\leq$0.05 in others), like those listed in Table 2. In these SNPs, observing multiple heterozygotes is informative for admixture as they occur at very low frequencies in the unadmixed contributor populations, e.g., the likelihood of two heterozygotes in SNPs with the above frequencies would be 1% in unadmixed populations, but 81% in an individual with parental co-ancestry from those contributing populations. Our analyses underlined the importance of uniparental markers when looking at patterns of admixture in a forensic sample, particularly contrasting inferred patrilineal and matrilineal ancestries in male DNA. The same applies to the extra information potentially provided by X-SNPs - with the single PIMA X marker adding extra detail. Since Minstead we have developed a 16 X-SNP set, which has been successfully incorporated into a massively parallel sequencing ancestry and FDP SNP panel [47]. Lastly, it has recently become evident that enlarged forensic SNP panels comprising several thousand loci provide the most accurate estimation of co-ancestry, even when individual SNPs are not selected as ancestry informative [48]. We see such panels as the future of ancestry analysis in forensic casework – SNP panels with distinct advantages for challenging human identification: IGG, distant-kinship analysis of missing persons or mixture de-convolution; and in parallel provide highly informative data on a person's ancestry.

CRediT authorship contribution statement

C. Phillips: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation; **A. Rodriguez:** Writing – review & editing, Methodology; **A. Gómez-Tato:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation; **J. Álvarez-Dios:** Writing – review & editing, Methodology, Formal analysis, Data curation; **M. Casares de Cal:** Writing – review & editing, Methodology, Formal analysis, Data curation; **A. Salas:** Writing – review & editing, Methodology, Formal analysis, Data curation; **Á. Carracedo:** Writing – review & editing, Formal analysis, Data curation; **M.V. Lareu:** Writing – review & editing, Methodology, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare they have no competing interests

Acknowledgements

The authors are indebted to the scientists who were working at the Forensic Genetics Unit of the University of Santiago de Compostela at the time of the genetic analyses made in support of the Operation Minstead investigations in 2008: Manuel Fondevila (ancestry SNP assay development and analyses), Yarimar Ruiz (SHEP assay development and analyses), and Vanessa Álvarez-Iglesias (*MC1R* sequencing). We also wish to thank Adam Auton for generously sharing the underlying data of the genetic cluster analyses made by the 1000 Genomes Project.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.fsigen.2026.103527.

References

[1] A. Salas, M. Richards, M.V. Lareu, R. Scozzari, A. Coppa, A. Torroni, V. Macaulay, A. Carracedo, The African diaspora: mitochondrial DNA and the Atlantic slave trade, Am. J. Hum. Genet. 74 (2004) 454–465.

[2] T. King, E.J. Parkin, G. Swinfield, F. Cruciani, R. Scozzari, A. Rosa, S.K. Lim, Y. Xue, C. Tyler-Smith, M.A. Jobling, Africans in Yorkshire? The deepest-rooting clade of the Y phylogeny within an English genealogy, Eur. J. Hum. Genet. 15 (2007) 288–293.

[3] A.L. Lowe, A. Urquhart, L.A. Foreman, I.W. Evett, Inferring ethnic origin by means of an STR profile, Forensic Sci. Int. 119 (2001) 17–22.

[4] C. Phillips, M. Gelabert-Besada, L. Fernandez-Formoso, M. García-Magariños, C. Santos, M. Fondevila, D. Ballard, D. Syndercombe Court, A. Carracedo, M. V. Lareu, "New turns from old STaRs": enhancing the capabilities of forensic short tandem repeat analysis, Electrophoresis 35 (2014) 3173–3187.

[5] T.N. Frudakis, Molecular Photofitting, Predicting ancestry and phenotype using DNA, Book, Academic Press 2008, ISBN 978-0-12-088492-6.

[6] J.Benn Torres, R.A. Kittles, A.C. Stone, Mitochondrial and Y Chromosome diversity in the English-speaking Caribbean, Ann. Hum. Genet. 71 (2007) 782–790.

[7] J. Benn-Torres, C. Bonilla, C.M. Robbins, L. Waterman, T.Y. Moses, W. Hernandez, E.R. Santos, F. Bennett, et al., Admixture and population stratification in African Caribbean populations, Ann. Hum. Genet. 72 (2008) 90–98.

[8] D.A. Bolnick, D. Fullwiley, T. Duster, R.S. Cooper, J.H. Fujimura, J. Kahn, J. S. Kaufman, J. Marks, et al., The science and business of genetic ancestry testing, Science 318 (2007) 399–400.

[9] C.J. Hoggart, E.J. Parra, M.D. Shriver, C. Bonilla, R.A. Kittles, D.G. Clayton, P. M. McKeigue, Control of confounding of genetic associations in stratified populations, Am. J. Hum. Genet. 72 (2003) 1492–1504.

[10] C.J. Hoggart, M.D. Shriver, R.A. Kittles, D.G. Clayton, P.M. McKeigue, Design and analysis of admixture mapping studies, Am. J. Hum. Genet. 74 (2004) 965–978.

[11] P. McKeigue, P. (2007). ADMIXMAP: A program to model admixture using marker genotype data [Online]. Available online at: ⟨http://homepages.ed.ac.uk/pmckeigue/admixmap/admixmap_manual.pdf⟩.

[12] J.K. Pritchard, M. Stephens, P. Donnelly, Inference of population structure using multilocus genotype data, Genetics 155 (2000) 945–959.

[13] N.A. Rosenberg, J.K. Pritchard, J.L. Weber, H.M. Cann, K.K. Kidd, L. A. Zhivotovsky, M.W. Feldman, Genetic structure of human populations, Science 298 (2002) 2381–2385.

[14] D.J. Lawson, G. Hellenthal, S. Myers, D. Falush, Inference of population structure using dense haplotype data, PLoS. Genet. 8 (2011) e1002453.

[15] The 1000 Genomes Project Consortium, A. Auton, L.D. Brooks, R.M. Durbin, E. P. Garrison, H.M. Kang, J.O. Korbel, J.L. Marchini, S. McCarthy, G.A. McVean, et al., A global reference for human genetic variation, Nature 526 (2015) 68–74.

[16] C. Phillips, A. Salas, J.J. Sánchez, M. Fondevila, A. Gómez-Tato, J. Álvarez-Dios, M. Calaza de Cal, D. Ballard, M.V. Lareu, Á. Carracedo, The SNP*for*ID Consortium, Inferring ancestral origin using a single multiplex assay of ancestry-informative marker SNPs, Forensic Sci. Int. Genet. 1 (2007) 273–280.

[17] C. Carvalho Gontijo, L.G. Porras-Hurtado, A. Freire-Aradas, M. Fondevila, C. Santos, A. Salas, J. Henao, C. Isaza, et al., PIMA: a population informative multiplex for the Americas, Forensic Sci. Int. Genet. 44 (2020) 102200.

[18] J.Z. Li, D.M. Absher, H. Tang, A.M. Southwick, A.M. Casto, S. Ramachandran, H. M. Cann, G.S. Barsh, M. Feldman, L.L. Cavalli-Sforza, R.M. Myers, Worldwide human relationships inferred from genome-wide patterns of variation, Science 319 (2008) 1100–1104.

[19] G. Evanno, S. Regnaut, J. Goudet, Detecting the number of clusters of individuals using the software STRUCTURE: a simulation study, Mol. Ecol. 14 (2005) 2611–2620.

[20] Y. Ruiz, C. Phillips, A. Gomez-Tato, J. Alvarez-Dios, M. Casares de Cal, R. Cruz, O. Maronas, J. Sōchtig, M. Fondevila, M.J. Rodriguez-Cid, A. Carracedo, M. V. Lareu, Further development of forensic eye color predictive tests, Forensic Sci. Int. Genet. 7 (2013) 28–40.

[21] O. Maroñas, C. Phillips, J. Sōchtig, A. Gomez-Tato, R. Cruz, J. Alvarez-Dios, M. Casares de Cal, Y. Ruiz, M. Fondevila, A. Carracedo, M.V. Lareu, Development of a forensic skin colour predictive test, Forensic Sci. Int. Genet. 13 (2014) 34–44.

[22] P. Sulem, D.F. Gudbjartsson, S.N. Stacey, A. Helgason, T. Rafnar, K.P. Magnusson, A. Manolescu, A. Karason, et al., Genetic determinants of hair, eye and skin pigmentation in Europeans, Nat. Genet. 39 (2007) 1443–1452.

[23] P. Sulem, D.F. Gudbjartsson, S.N. Stacey, A. Helgason, T. Rafnar, M. Jakobsdottir, S. Steinberg, S.A. Gudjonsson, et al., Two newly identified genetic determinants of pigmentation in Europeans, Nat. Genet. 40 (2008) 835–837.

[24] J. Han, P. Kraft, H. Nan, Q. Guo, C. Chen, A. Qureshi, S.E. Hankinson, F.B. Hu, et al., A Genome-wide association study identifies novel alleles associated with hair color and skin pigmentation, PLoS. Genet. 4 (2008) e1000074.

[25] R.A. Sturm, D.L. Duffy, Z.Z. Zhao, F.P. Leite, M.S. Stark, N.K. Hayward, N. G. Martin, G.W. Montgomery, A single SNP in an evolutionary conserved region within intron 86 of the HERC2 gene determines human blue-brown eye color, Am. J. Hum. Genet. 82 (2008) 424–431.

[26] M. Kayser, F. Liu, A. Cecile, J.W. Janssens, F. Rivadeneira, O. Lao, K. van Duijn, M. Vermeulen, et al., Three genome-wide association studies and a linkage analysis identify HERC2 as a human iris color gene, Am. J. Hum. Genet. 82 (2008) 411–423.

[27] H. Eiberg, J. Troelsen, M. Nielsen, A. Mikkelsen, J. Mengel-From, K. Kjaer, Blue eye color in humans may be caused by a perfectly associated founder mutation in a regulatory element located within the HERC2 gene inhibiting OCA2 expression, Hum. Mol. Genet. 123 (2008) 177–187.

[28] H.L. Norton, R.A. Kittles, E. Parra, P. McKeigue, X. Mao, K. Cheng, V.A. Canfield, D. G. Bradley, B. McEvoy, M.D. Shriver, Genetic evidence for the convergent evolution of light skin in Europeans and East Asians, Mol. Biol. Evol. 24 (2007) 710–722.

[29] R.A. Sturm, Molecular genetics of human pigmentation diversity, Hum. Mol. Genet. 18 (2008) R9–R17.

[30] R.P. Stokowski, P.V. Pant, T. Dadd, A. Fereday, D.A. Hinds, C. Jarman, W. Filsell, R.S. Ginger, M.R. Green, F.J. van der Ouderaa, D.R. Cox, A genomewide association study of skin pigmentation in a South Asian population, Am. J. Hum. Genet. 81 (2007) 1119–1132.

[31] J.C. Garcia-Borrón, B.L. Sánchez-Laorden, C. Jiménez-Cervantes, Melanocortin-1 receptor structure and functional regulation, Pigment. Cell. Res. 18 (2005) 393–410.

[32] A.E. Grimes, P.J. Noake, L. Dixon, A. Urquhart, Sequence polymorphism in the human melanocortin 1 receptor gene as an indicator of the red hair phenotype, Forensic Sci. Int. 122 (2001) 124–129.

[33] R.M. Harding, J.Brown Tomlinson, A.J. Ray, K. Wakamatsu, J.L. Rees, C. A. McKenzie, Phenotypic expression of melanocortin-1 receptor mutations in Black Jamaicans, J. Invest. Dermatol. 121 (2003) 207–208.

[34] M.R. Gerstenblith, A.M. Goldstein, M.C. Fargnoli, K. Peris, M.T. Landi, Comprehensive evaluation of allele frequency differences of MC1R variants across populations, Hum. Mut. 28 (2007) 495–505.

[35] G. Stevens, M. Ramsay, T. Jenkins, Oculocutaneous albinism (OCA2) in sub-Saharan Africa: distribution of the common 2.7-kb P gene deletion mutation, Hum. Genet. 99 (1997) 523–527.

[36] P. Manga, J. Kromberg, A. Turner, T. Jenkins, M. Ramsay, Southern Africa, brown oculocutaneous albinism (BOCA) maps to the OCA2 locus on chromosome 15q: P-gene mutations identified, Am. J. Hum. Genet. 68 (2001) 782–787.

[37] M. Coquet, N. Terrado-Ortuño, Forensic DNA phenotyping: privacy breach, bias reification and the pitfalls of abstract assessment of rights, Int. J. Police Sci. Manag. 25 (2023) 262–279.

[38] N. Terrado-Ortuño, P. May, Forensic DNA phenotyping: a review on SNP panels, genotyping techniques, and prediction models, Forensic Sci. Res. 10 (2024) owae013.

[39] C. Phillips, M. de la Puente, The analysis of ancestry with small-scale forensic panels of genetic markers, Emerg. Top. Life Sci. 5 (2021) 443–453.

[40] C. Phillips, L. Prieto, M. Fondevila, A. Salas, A. Gomez-Tato, J.A. Alvarez-Dios, A. Alonso, A. Blanco-Verea, M. Brión, M. Montesino, et al., Ancestry analysis in the 11-M Madrid bomb attack investigation, PLoS. One 4 (2009) e6583.

[41] C. Phillips, Forensic genetic analysis of bio-geographical ancestry, Forensic Sci. Int. Genet. 18 (2015) 49–65.

[42] V. Pereira, A. Freire-Aradas, D. Ballard, C. Børsting, V. Diez, P. Pruszkowska-Przybylska, J. Ribeiro, N.M. Achakzai, A. Aliferi, O. Bulbul, et al., Development and validation of the EUROFORGEN NAME (North African and Middle Eastern) ancestry panel, Forensic Sci. Int. Genet. 42 (2019) 260–267.

[43] I. Halder, M. Shriver, M. Thomas, J.R. Fernandez, T. Frudakis, A panel of ancestry informative markers for estimating individual bio-geographical ancestry and admixture from four continents: utility and applications, Hum. Mutat. 29 (2008) 648–658.

[44] P.A. Bell, S. Chaturvedi, C.A. Gelfand, C.Yu Huang, M. Kochersperger, R. Kopla, F. Modica, M. Pohl, et al., SNPstream UHT: ultra-high throughput SNP genotyping for pharmacogenomics and drug discovery, Biotechniques 74 (2002) 76–77.

[45] W. Branicki, A. Pisarek-Pacek, K. Marszałek, A. Jarosz, M. Kukla-Bartoszek, M. Zubańska, A. Bronikowska, K. Węgrzyn, et al., Investigation of the MC1R gene sequence variation using Oxford Nanopore Sequencing, Electrophoresis 46 (2025) 727–731.

[46] C. Santos, C. Phillips, A. Gomez-Tato, J. Alvarez-Dios, Á. Carracedo, M.V. Lareu, Inference of ancestry in forensic analysis II: analysis of genetic data, Methods Mol. Biol. 1420 (2016) 255–285.

[47] J. Ruiz-Ramírez, M. de la Puente, C. Xavier, A. Ambroa-Conde, J. Álvarez-Dios, A. Freire-Aradas, A. Mosquera-Miguel, A. Ralf, et al., Development and evaluations of the ancestry informative markers of the VISAGE enhanced tool for appearance and ancestry, Forensic Sci. Int. Genet. 64 (2023) 102853.

[48] M. de la Puente, L. Casanova-Adán, J. González-Bao, J. Pardo-Seco, A. Mosquera-Miguel, A. Ambroa-Conde, J. Ruiz-Ramírez, A. Cabrejas-Olalla, et al., Evaluating the effect of marker panel sizes on estimation of bio-geographical co-ancestry proportions, Forensic Sci. Int. Genet. 78 (2025) 103275.